

1a) In the intersection points, $\begin{pmatrix} t \\ 0 \\ 2t-t^2 \end{pmatrix} = \begin{pmatrix} x \\ y \\ x^2+y^2 \end{pmatrix}$,

i.e. $x=t$, $y=0$, $z=x^2+y^2=t^2$,

but also $z=2t-t^2$

$$\rightarrow t^2 = 2t - t^2$$

$$2t^2 = 2t$$

$$t=0 \quad \text{or} \quad t=1$$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

b) The sphere with centre (a, b, c) and radius r is given by

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

$$x^2 + y^2 + z^2 - 8x + 2y + 6z + 1 = 0$$

$$\rightarrow (x-4)^2 - 16 + (y+1)^2 - 1 + (z+3)^2 - 9 + 1 = 0$$

$$\rightarrow (x-4)^2 + (y+1)^2 + (z+3)^2 = 25$$

$$\rightarrow \text{Centre} = (4, -1, -3)$$

$$\text{radius} = 5$$

2a) On the curve of intersection, we have

$$y = x^2 \quad \text{and} \quad z = 4x^2 + y^2 = 4x^2 + x^4$$

$$\Rightarrow \text{If we set } x=t, \text{ then } y=t^2$$

$$\text{and } z = 4t^2 + t^4$$

$$\Rightarrow t \mapsto \begin{pmatrix} t \\ t^2 \\ 4t^2+t^4 \end{pmatrix} \text{ is a parametrisation of the curve.}$$

$$b) F'(t) = 2t \underline{i} + 2t \underline{j} + \underline{k}$$

$$F(t) = -\underline{i} + \underline{j} + \underline{k} \rightarrow t=0$$

$$F'(0) = \underline{k}$$

$\rightarrow$ Tangent line at $-\underline{i} + \underline{j} + \underline{k}$ is given by

$$t \mapsto t F'(0) + F(0)$$

$$= -\underline{i} + \underline{j} + (t+1) \underline{k}$$

3) The particles collide if $r_1(t) = r_2(t)$ for some $t \geq 0$.

$$\Rightarrow t^2 = 4t - 3$$

$$7t - 12 = t^2$$

$$t^2 = 5t - 6$$

From first and third equation, we get

$$5t - 6 = 4t - 3$$

$$\Rightarrow t = 3$$

$\Rightarrow$ If the particles collide, they do so at $t=3$.

$$\text{Check: } r_1(3) = 9\mathbf{i} + 9\mathbf{j} + 9\mathbf{k}$$

$$r_2(3) = 9\mathbf{i} + 9\mathbf{j} + 9\mathbf{k}$$

$\Rightarrow$ They do collide.

4 a) Substitute the coordinates of $F(t)$ into the equation $4x + 5y - 2z = 18$:

$$4(2+3t) + 5(-4t) - 2(5+t) = 18$$

$$8 + 12t - 20t - 10 - 2t = 18$$

$$-10t = 20$$

$$t = -2$$

$\Rightarrow$ The point of intersection is

$$F(-2) = -4\mathbf{i} + 8\mathbf{j} + 3\mathbf{k}$$

b) In general, the equation of a plane looks like $ax + by + cz = d$.

Substituting the points $\mathbf{i}$, $2\mathbf{j}$ and $3\mathbf{k}$ gives

$$a = d$$

$$2b = d$$

$$3c = d$$

$\Rightarrow$ Taking $d=6$, we get $a=6$, $b=3$, $c=2$

$$\Rightarrow 6x + 3y + 2z = 6$$

4 c) Consider the normal vectors

$$\underline{n}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad \underline{n}_2 = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$$

The planes are perpendicular/parallel $\Leftrightarrow$
 $\underline{n}_1$ and $\underline{n}_2$ are perpendicular/parallel.

$\underline{n}_1$ and $\underline{n}_2$ are not parallel $\Rightarrow$ the planes are not parallel.

$$\underline{n}_1 \cdot \underline{n}_2 = 2 - 3 - 4 = -5 \neq 0 \\ \Rightarrow \text{not perpendicular.}$$

If θ is the angle between the planes
(= the angle between $\underline{n}_1$ and $\underline{n}_2$), then

$$\cos \theta = \frac{\underline{n}_1 \cdot \underline{n}_2}{\|\underline{n}_1\| \|\underline{n}_2\|}$$

$$\|\underline{n}_1\| = \sqrt{1^2 + 1^2 + (-1)^2} = \sqrt{3}, \quad \|\underline{n}_2\| = \sqrt{2^2 + (-3)^2 + 4^2} = \sqrt{29}$$

$$\Rightarrow \cos \theta = \frac{-5}{\sqrt{3} \cdot \sqrt{29}}$$

$$\Rightarrow \cos \theta \approx -2.137, \text{ rad} \approx -122.4^\circ$$

d) Any plane parallel to $x + 4y - 3z = 1$
has an equation of the form $x + 4y - 3z = d$.
To find d , we substitute the point $2\mathbf{i} + \mathbf{j}$:
 $2 + 4 = d \Rightarrow d = 6$

$$\Rightarrow x + 4y - 3z = 6.$$

5a) $F'(t) = 2t \underline{i} + 5 \underline{j} + (2t - 16) \underline{k}$

$$\|F'(t)\| = \sqrt{(2t)^2 + 5^2 + (2t - 16)^2}$$

$$= \sqrt{8t^2 - 64t + 281}$$

Want to minimise $\sqrt{8t^2 - 64t + 281}$,

or equivalently, minimise $8t^2 - 64t + 281$.

$$\frac{d}{dt} (8t^2 - 64t + 281) = 16t - 64 = 0$$

$\Rightarrow t = 4$ is the time when the speed is the least.

b) Let $\underline{v}(t) = \begin{pmatrix} v_1(t) \\ v_2(t) \\ v_3(t) \end{pmatrix}$ be the velocity vector and let c be the (constant) speed.

$$\sqrt{v_1(t)^2 + v_2(t)^2 + v_3(t)^2} = c$$

$$v_1(t)^2 + v_2(t)^2 + v_3(t)^2 = c^2$$

Differentiate:

$$2v_1(t)v_1'(t) + 2v_2(t)v_2'(t) + 2v_3(t)v_3'(t) = 0.$$

Note that the left-hand side is equal to $2 \underline{v}(t) \cdot \underline{v}'(t)$

Hence, $\underline{v}(t) \cdot \underline{v}'(t) = 0$, i.e. the velocity and acceleration vectors are perpendicular.

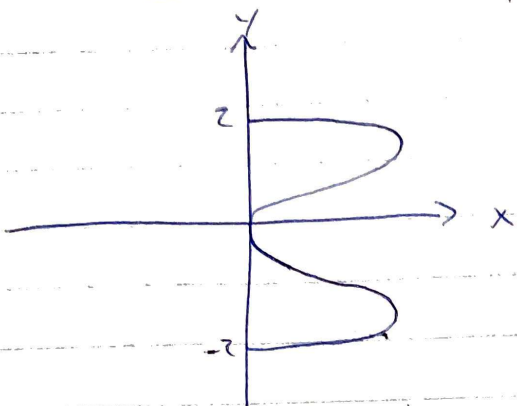
6) We can find a parametrisation of the ellipsoid by taking a parametrisation of the unit sphere and multiplying by the half-axes. The sphere is parametrised by

$$(\phi, \theta) \mapsto \begin{pmatrix} \sin \phi \cos \theta \\ \sin \phi \sin \theta \\ \cos \phi \end{pmatrix}$$

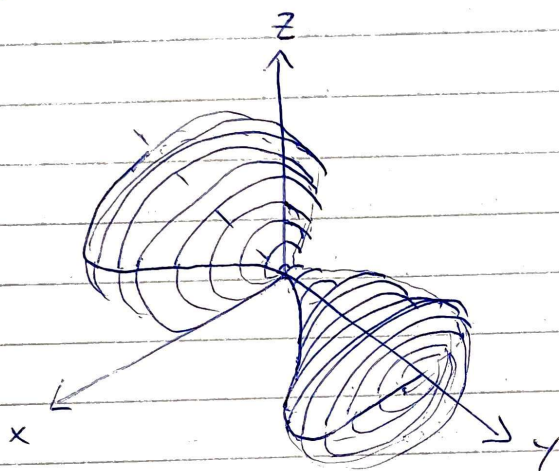
Hence, the ellipsoid is parametrised by

$$(\phi, \theta) \mapsto \begin{pmatrix} 2 \sin \phi \cos \theta \\ \sin \phi \sin \theta \\ 3 \cos \phi \end{pmatrix}$$

7) The curve $x = 4y^2 - y^4$ looks like



If we rotate this about the y -axis, we get



Fix $y=t$. The part of the surface where $y=t$ is a circle with radius $4t^2 - t^4$, so it's parametrised by

$$\theta \mapsto (4t^2 - t^4) \cos \theta, (4t^2 - t^4) \sin \theta.$$

Hence, the surface of revolution is parametrised by

$$(t, \theta) \mapsto \begin{pmatrix} (4t^2 - t^4) \cos \theta \\ t \\ (4t^2 - t^4) \sin \theta \end{pmatrix}$$

Question 8

i) On the y -axis, $f(x, y) = f(0, y) = 0$.

Hence, $\lim_{y \rightarrow 0} f(0, y) = 0$.

On the curve $x = y^2$, $f(x, y) = \frac{y^4}{2y^4} = \frac{1}{2}$.

Hence, $\lim_{y \rightarrow 0} f(y^2, y) = \frac{1}{2}$.

Since these two limits are different,

$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^2}$ doesn't exist.

ii) For any point (x, y) , let r be its distance to the origin.

Then $r^2 = x^2 + y^2$,

$|x| \leq r$ and $|y| \leq r$.

Hence, $\left| \frac{3x^2y}{x^2+y^2} \right| \leq \frac{3r^3}{r^2} = 3r$

$\Rightarrow \left| \frac{3x^2y}{x^2+y^2} \right| \rightarrow 0$ as $r \rightarrow 0$

(i.e. as $(x, y) \rightarrow (0, 0)$)

$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2+y^2} = 0$

Question 9

i) $f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$

$= \lim_{h \rightarrow 0} \frac{(x+h)y^2 - (x+h)^2 y - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} \frac{xy^2 + hy^2 - x^2 y - 2xhy - h^2 y - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} y^2 - 2xy - hy = y^2 - 2xy$

ii) $f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$

$= \lim_{h \rightarrow 0} \frac{x(y+h)^2 - x^2(y+h) - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} \frac{xy^2 + 2xyh + xh^2 - x^2 y - x^2 h - xy^2 + x^2 y}{h}$

$= \lim_{h \rightarrow 0} 2xy + xh - x^2 = 2xy - x^2$

Question 9

$$\begin{aligned}
 f_x(x, y) &= \lim_{h \rightarrow 0} \frac{\frac{x+h}{x+h+y^2} - \frac{x}{x+y^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{(x+h)(x+y^2) - x(x+h+y^2)}{(x+h+y^2)(x+y^2)h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + xy^2 + xh + y^2h - \cancel{x^2} - xh - xy^2}{(x+h+y^2)(x+y^2)h} \\
 &= \lim_{h \rightarrow 0} \frac{y^2}{(x+h+y^2)(x+y^2)} = \frac{y^2}{(x+y^2)^2}
 \end{aligned}$$

$$\begin{aligned}
 f_y(x, y) &= \lim_{h \rightarrow 0} \frac{\frac{x}{x+(y+h)^2} - \frac{x}{x+y^2}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{x(x+y^2) - x(x+(y+h)^2)}{(x+(y+h)^2)(x+y^2)h} \\
 &= \lim_{h \rightarrow 0} \frac{\cancel{x^2} + xy^2 - \cancel{x^2} - x(y^2 + 2yh + h^2) - xh^2}{(x+(y+h)^2)(x+y^2)h} \\
 &= \lim_{h \rightarrow 0} \frac{-2xy - xh}{(x+(y+h)^2)(x+y^2)} = \frac{-2xy}{(x+y^2)^2}
 \end{aligned}$$

Question 10

$$\begin{aligned}
 i) \quad \frac{\partial}{\partial x} (yz) &= \frac{\partial}{\partial x} \ln(x+z) \\
 y \frac{\partial z}{\partial x} &= \frac{1}{x+z} \left(1 + \frac{\partial z}{\partial x}\right)
 \end{aligned}$$

$$\left(y - \frac{1}{x+z}\right) \frac{\partial z}{\partial x} = \frac{1}{x+z}$$

$$\frac{\partial z}{\partial x} = \frac{\frac{1}{x+z}}{y - \frac{1}{x+z}} = \frac{1}{y(x+z) - 1}$$

$$\frac{\partial}{\partial y} (yz) = \frac{\partial}{\partial y} \ln(x+z)$$

$$z + y \frac{\partial z}{\partial y} = \frac{1}{x+z} \frac{\partial z}{\partial y}$$

$$\left(y - \frac{1}{x+z}\right) \frac{\partial z}{\partial y} = -z$$

$$\frac{\partial z}{\partial y} = \frac{-z}{y - \frac{1}{x+z}} = \frac{-z(x+z)}{y(x+z) - 1}$$

$$ii) \frac{\partial}{\partial x} (x - z) = \frac{\partial}{\partial x} \arctan(yz)$$

$$1 - \frac{\partial z}{\partial x} = \frac{1}{1+(yz)^2} \cdot y \frac{\partial z}{\partial x}$$

$$\left(1 + \frac{y}{1+(yz)^2}\right) \frac{\partial z}{\partial x} = 1$$

$$\frac{\partial z}{\partial x} = \frac{1}{1 + \frac{y}{1+(yz)^2}} = \frac{1+(yz)^2}{1+(yz)^2 + y}$$

$$\frac{\partial}{\partial y} (x - z) = \frac{\partial}{\partial y} \arctan(yz)$$

$$-\frac{\partial z}{\partial y} = \frac{1}{1+(yz)^2} \left(z + y \frac{\partial z}{\partial y}\right)$$

$$\left(1 + \frac{y}{1+(yz)^2}\right) \frac{\partial z}{\partial y} = \frac{-z}{1+(yz)^2}$$

$$\frac{\partial z}{\partial y} = \frac{\frac{-z}{1+(yz)^2}}{1 + \frac{y}{1+(yz)^2}} = \frac{-z}{1+(yz)^2 + y}$$

Question 11

$$i) f_x(x, y) = \sin(x+zy) + x \cos(x+zy)$$

$$f_{xy}(x, y) = z \cos(x+zy) - zx \sin(x+zy)$$

$$f_y(x, y) = zx \cos(x+zy)$$

$$f_{yx}(x, y) = z \cos(x+zy) - zx \sin(x+zy)$$

Same

$$ii) f_x(x, y) = 4x^3 y^2 - 2y^5$$

$$f_{xy}(x, y) = 8x^3 y - 10y^4$$

$$f_y(x, y) = 2x^4 y - 10xy^4$$

$$f_{yx}(x, y) = 8x^3 y - 10y^4$$

Same

Question 12

$$i) f_x(x, y) = \frac{3}{3x+5y}$$

$$f_y(x, y) = \frac{5}{3x+5y}$$

$$f_{xx}(x, y) = \frac{-3}{(3x+5y)^2}$$

$$f_{xy}(x, y) = f_{yx}(x, y) = \frac{-15}{(3x+5y)^2}$$

$$f_{yy}(x, y) = \frac{-25}{(3x+5y)^2}$$

$$i) \quad z_x = \frac{x+y-x}{(x+y)^2} = \frac{y}{(x+y)^2}$$

$$z_y = \frac{-x}{(x+y)^2}$$

$$z_{xx} = \frac{-2y}{(x+y)^3}$$

$$z_{xy} = z_{yx} = \frac{(x+y)^2 - 2y(x+y)}{(x+y)^4} = \frac{x+y-2y}{(x+y)^3} = \frac{x-y}{(x+y)^3}$$

$$z_{yy} = \frac{2x}{(x+y)^3}$$

Question 13

$$i) \quad f_t = -cx^2 e^{-ct}$$

$$f_{tt} = c^2 x^2 e^{-ct}$$

$$f_{ttt} = -c^3 x^2 e^{-ct}$$

$$f_{tx} = -2cx e^{-ct}$$

$$f_{txx} = -2c e^{-ct}$$

$$ii) \quad f_x = -4 \sin(4x+3y+2z)$$

$$f_{xy} = -12 \cos(4x+3y+2z)$$

$$f_{xyz} = 24 \sin(4x+3y+2z)$$

$$\Rightarrow f_{zyx}$$

Question 14

$$i) \quad \text{The ellipse is given by } 4x^2 + 8 + z^2 = 16$$

$$\Rightarrow 4x^2 + z^2 = 8$$

To find the slope of the tangent line, we use implicit differentiation:

$$8x + 2z \frac{dz}{dx} = 0$$

$$\frac{dz}{dx} = -\frac{8x}{2z} = -4 \frac{x}{z}$$

$\Rightarrow$ The tangent line at the point $(1, 2, 2)$ has slope $-4 \cdot \frac{1}{2} = -2$.

$\Rightarrow$ The equation of the tangent line is

$$z - 2 = -2(x - 1)$$

$$z = -2x + 4$$